Remote Sensing Div. B Answer Key
Tiebreakers: 15, 21, 31, 46, 60, 64

1. Remote sensing is obtaining information about an object from a distance, usually using satellites or aircrafts to detect and measure reflected and emitted radiation, OR a set of techniques used to collect information about an area or object (a target) without being in physical contact with that target.
2. A-4, B-5, C-1, D-2, E-3
3. B
4. D
5. low-Earth orbit (approximately 160/200 to 2,000 km above Earth), medium-Earth orbit (approximately 2,000 to 35,500/35,786 km above Earth), and high-Earth orbit (above 35,500/35786 km above Earth). Give one point for each orbit correct, half a point if one of the min or maxs are wrong, the / means either is acceptable different sources say different things.
6. C
7. 1 for "SAR/microwave (C/L-band)", +1 cloud-penetration, +1 day/night active illumination, +1 water = low backscatter.
8. D
9. (.62-.18)/(.62+.18) = .55. Healthy, actively photosynthesizing vegetation (typical: bare soil ~0–0.2, moderate 0.3–0.5, vigorous 0.5–0.8)
10. B
11. Geostationary orbit (GEO), because the satellite stays over the same point on Earth and provides continuous, high-frequency images of the same area.
12. A radar altimeter, because it measures the time delay of microwave pulses bouncing off the ocean surface. Optical sensors can't penetrate clouds and cannot directly measure distance to the surface.
13. Pushbroom scanners collect an entire line at once using a detector array, so they have no moving mirror, produce less geometric distortion, and allow longer dwell time per pixel (better SNR).
14. South 32 °West
15. They are created by pivot irrigation systems that water crops in circular patterns, leaving the corners of fields dry. The green areas show healthy irrigated crops, while the brown areas are rangeland or unwatered zones.
16. Water vapor in the atmosphere blocks out most of the light in those wavelengths, so the satellite can't measure anything there.
17. Answer: Landsat TM (Thematic Mapper) was the first to include a thermal infrared band (+2 pts), operating around 10.4–12.5 μm (Band 6) (+2 pts).

18. OLI has more bands, including a coastal/aerosol band and SWIR bands, which help identify land and water types better.
19. SWIR bands (e.g., OLI Bands 6 and 7) (+1 pt), because snow has low reflectance in SWIR while clouds remain highly reflective, allowing spectral separation that is not possible in visible wavelengths (+2 pts).
20. Because thermal bands measure heat energy the Earth gives off, not sunlight it reflects, and that energy is in the long-wavelength infrared.
21. B
22. Farmland = green/blue (high ETa → vegetation → active transpiration); urban = tan (low ETa → impervious surfaces).
23. Cooler surfaces = high ET (water evaporates, cooling surface); warmer = low ET (dry soil or pavement).
24. Plants convert absorbed solar energy into latent heat (evaporation/transpiration), using energy to move water rather than heat surfaces.
25. It's detailed enough to detect field-scale variations in water use while still covering large regions efficiently.
26. They identify which fields or zones are using more or less water, guiding targeted irrigation and conserving resources.
27. It provides near-real-time tracking of surface temperature and ETa, enabling quicker detection of crop stress, urban heat, or hydrologic change.
28. 1. Ridge, 2. Valley, 3. Canyon
29. B
30. A
31. C
32. Moderate Resolution Imaging Spectroradiometer
33. 1, 4, 3
34. Reflected or emitted
35. B
36. Albedo is a term referring to the measure of reflectivity of the Earth's surface
37. A: Unsupervised classification B: Supervised Classification
38. Geographic and Projected
39. EOS AM-1
40. LIDAR
41. Backscatter, "hit" and return of a laser pulse
42. A
43. Aircraft condensation trail (contrail)
44. Low Earth orbit satellite
45. Multispectral Scanner (MSS) and Thematic Mapper (TM)
46. Panchromatic: single channel detector sensitive to radiation in a broad range, Multispectral: multichannel detector with a few spectral bands,

Superspectral/Ultraspectral: many spectral bands (>10), narrow ranges in bands, Hyperspectral: lots of contiguous spectral bands, >100.

47. B
48. Milutin Milankovitch
49. D
50. 50 km
51. B
52. C
53. short-term, natural fluctuations in the climate system that make it difficult to distinguish long-term climate signals
54. C
55. A
56. B
57. Operational Land Imager
58. Push broom scanner / Along-track scanner
59. Landsat 8
60. In the north, temperatures are colder, and the water is fresher (less saline) and shallower. Fresh water freezes at a higher temperature than salty water, and shallower areas allow water to cool more readily throughout the water column than in the deeper parts. Thus, the northern Caspian Sea facilitates freezing more than the south.
61. B
62. Because tree canopies cause multiple scattering and depolarization, creating strong cross-polarized backscatter. Smooth water reflects energy away from the radar, so little cross-polarized signal returns.
63. 2^12, or 4096 levels
64. UTM uses the Transverse Mercator projection; it minimizes distortion within each 6-degree longitudinal zone, giving accurate distance and area measurements for regional satellite scenes.